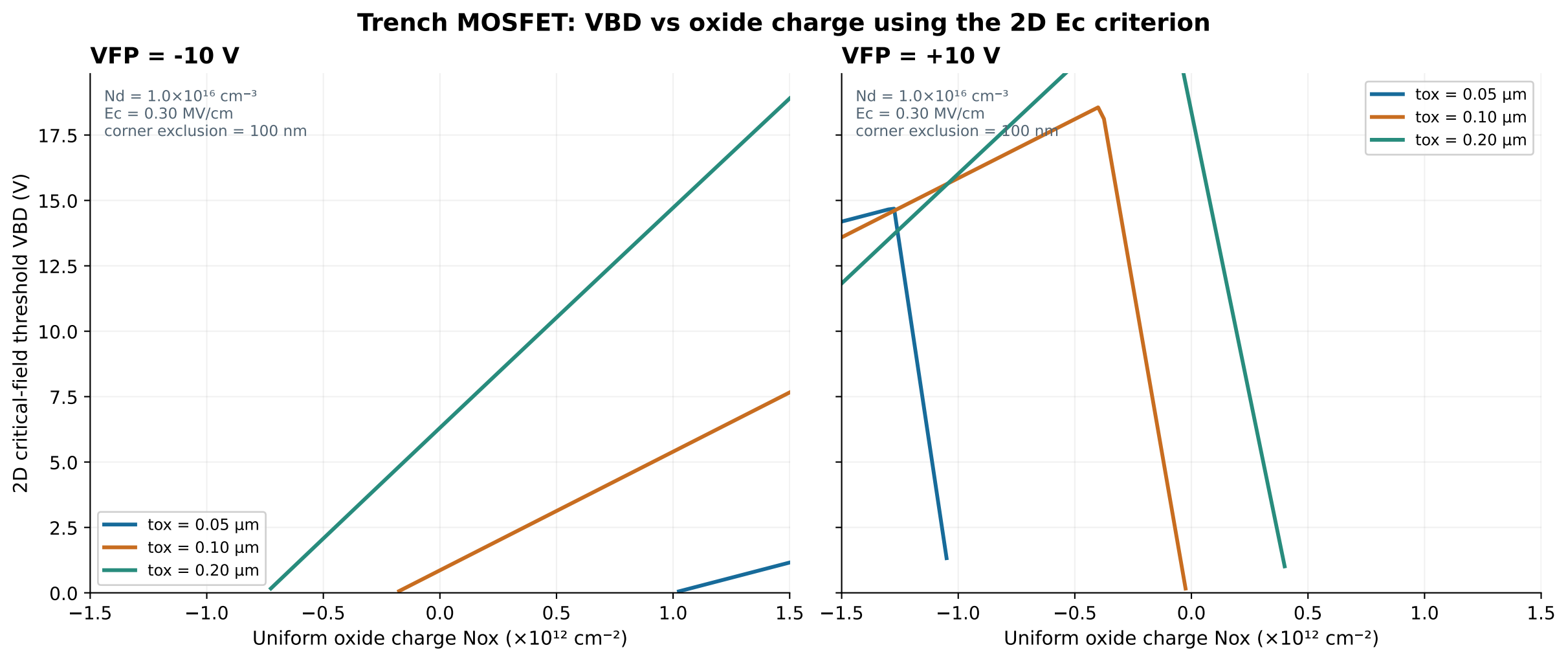
Trench MOSFET: VBD vs oxide charge using the 2D Ec criterion
VFP = -10 V
VFP = +10 V
Nd = 1.0×10¹⁶ cm⁻³
Ec = 0.30 MV/cm
corner exclusion = 100 nm
tox = 0.05 µm
tox = 0.10 µm
tox = 0.20 µm
2D critical-field threshold VBD (V)
Uniform oxide charge Nox (×10¹² cm⁻²)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5